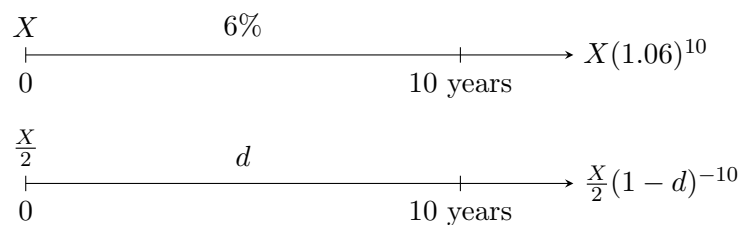


Problem 8.18

A deposit of X is made into a fund which pays an annual effective interest rate of 6% for 10 years. At the same time, $\frac{X}{2}$ is deposited into another fund which pays an annual effective rate of discount of d for 10 years. The amounts of interest earned over the 10 years are equal for both funds. Calculate d .

Solution

The two deposits and their accumulated values can be represented as follows:



The interest earned by the first fund is

$$X(1.06)^{10} - X = X((1.06)^{10} - 1).$$

For a fund with annual effective discount rate d , the accumulation factor is

$$\frac{1}{1-d} = (1-d)^{-1}.$$

Thus, the interest earned by the second fund is

$$\frac{X}{2}(1-d)^{-10} - \frac{X}{2} = \frac{X}{2}((1-d)^{-10} - 1).$$

Since the amounts of interest earned are equal,

$$X((1.06)^{10} - 1) = \frac{X}{2}((1-d)^{-10} - 1).$$

Canceling X and solving for d ,

$$\begin{aligned}2((1.06)^{10} - 1) &= (1 - d)^{-10} - 1, \\2((1.06)^{10} - 1) + 1 &= (1 - d)^{-10}, \\1 - d &= \left(\frac{1}{2((1.06)^{10} - 1) + 1} \right)^{1/10}, \\d &= 1 - \left(\frac{1}{2((1.06)^{10} - 1) + 1} \right)^{1/10}, \\d &\approx 0.09048.\end{aligned}$$

Therefore,

$$\boxed{d \approx 9.048\%}.$$